

# AI & Climate



# AI & Climate: Introduction and Motivation

- **Motivation:** Climate change is a critical issue affecting everyone
- **The Energy Cost of AI:** Training and deploying large models requires massive power.
- **GPT-3:** 1287 mWh, generated 552 tons of CO<sub>2</sub><sup>1</sup>



# Opportunities: Analyzing & Predicting Climate AI as a Superior Tool

## Data Processing

... can analyze massive amounts of climate data much faster than humans

## Pattern Recognition

... recognizes patterns in weather systems and can predict environmental changes months in advance

## Disaster Mitigation

... helps scientists understand complex climate processes, allows for earlier hurricane preparation, saving lives during extreme weather events <sup>2</sup>

## Monitoring

... tracks deforestation and human activity patterns related to the climate <sup>3</sup>



# Opportunities: Energy Optimization Smarter Consumption

- **Smart Grids:** predicting exactly when and where energy is needed
- **Building Efficiency:** Intelligent heating and cooling controls can reduce energy waste in buildings
- **Renewable Integration:** Predicts when solar or wind sources will produce the most energy, making green energy sources more reliable and efficient

# Risks: The Environmental Cost of AI

## Resource Consumption

### Energy Use

Large AI models and data centers consume huge amounts of energy, often derived from fossil fuels, increasing CO2 emissions <sup>5</sup>

### Water Scarcity

Data centers require massive amounts of water for cooling <sup>4</sup>

### Statistic

A typical data center uses around 300,000 gallons of water daily <sup>4</sup>

### Future Projection

Water use for AI could grow by up to 870%, threatening regions already facing water shortages <sup>3</sup>

### Reliability Risks

Reliance on bad data can lead to wrong predictions and delayed responses during disasters <sup>2</sup>

# Case Study: Pakistan (The Challenge)

## The Impact of Climate Disasters

### Recent History:

- **2022 Floods:** Unexpected monsoon water levels caused approx. 1,800 deaths <sup>6</sup>
- **2025 Floods:** Repeated events caused 800 deaths and significant economic / infrastructure loss <sup>7</sup>

### Systemic Failures:

- **Communication Gap:** Warnings were too generic (district-level), causing citizens to underestimate the danger in their specific neighborhoods
- **Aid Mismanagement:** Lack of communication between aid providers and affected areas led to uneven distribution and mismatched supplies



# Case Study: Pakistan (AI Improving Disaster Response Solutions)



## Precision Warnings

AI using real-time satellite imagery and rain data can predict specific flood zones, giving locals time to evacuate and protect assets



## Post-Flood Management

AI helps filter and predict upcoming issues, such as infection spreads (e.g., malaria), enabling the government to prepare timely responses.



## Supply Optimization

AI can filter supply needs based on geographical location, a task difficult to perform manually

# Case Study: Germany

## (Government Initiatives)

### "AI Lighthouse Projects"

- **Funding:** Since 2019, Germany has funded 53 projects focused on AI for climate and the environment <sup>8</sup>
- **Application Examples:**
  - Forestry: Using AI to determine which tree species to plant for climate resilience
  - Recycling: Optimizing waste sorting to increase plastic recycling rates
  - Transport: Optimizing transport systems to reduce CO2 emissions <sup>9</sup>
- **Monitoring:** The government actively monitors AI's environmental impact to minimize its resource consumption.



# Conclusion & Future Outlook

## Balancing Potential and Risk

- **Summary:** AI is a powerful tool for fighting climate change (better warnings, smarter energy) but brings its own environmental risks (energy/water consumption)
- **The Goal for 2040:** create clean, resource-efficient AI systems that minimize energy and water consumption, reduce the risks and consequences of climate change, and support the expansion and intelligent use of renewable energy sources

# Sources and AI Tools

- Sources

- 1: <https://news.mit.edu/2025/explained-generative-ai-environmental-impact-0117>
- 2: <https://www.npr.org/2025/11/29/nx-s1-5604290/atlantic-hurricane-season-forecast-ai-google>
- 3: <https://www.nature.com/articles/s41598-025-23617-4>
- 4 : <https://www.brookings.edu/articles/ai-data-centers-and-water/>
- 5 : <https://hai.stanford.edu/ai-index>
- 6: <https://www.worldbank.org/en/news/press-release/2022/10/28/pakistan-flood-damages-and-economic-losses-over-usd-30-billion-and-reconstruction-needs-over-usd-16-billion-new-assessment#:~:text=The%20floods%20affected%2033%20million,The%20crisis%20thus%20risks%20having>
- 7: <https://tribune.com.pk/story/2563997/can-ai-predict-next-climate-disaster-in-pakistan>
- 8: <https://www.bundesumweltministerium.de/en/topics/digitalisation/overview-digitalisation/our-funding-initiative-ai-lighthouse-projects-for-the-environment-climate-nature-and-resources>
- 9: <https://www.bundesumweltministerium.de/themen/digitalisierung/kuenstliche-intelligenz/kuenstliche-intelligenz-fuer-umwelt-und-klima>

- AI Tools

- Slide Generation: <https://gamma.app/de>
- Wording assistance: ChatGPT